

Mathematics A

Unit 2 • Chapter OL-T22 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

2 classified items • 8 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T22 • Expertise Q16+ • 2 items • 8 marks

Chapter	Items	Marks	Starts
01. OL-T22 Rearranging Formulas	2	8	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Rearranging Formulas

Unit 2 • Expertise • 8 marks • 2 item(s)

June 2025 | 4MA1-2H Q16 | 4 marks

Single squared/powered subject

January 2020 | 4MA1-2H Q16 | 4 marks

Subject in the denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

$$I = 8I'$$

(Total for Question 16 is 4 marks)

WORKING SPACE

Make t the subject**4 marks****Clear the denominator**

$$c(7 - 8t^2) = t^2 + 3 \Rightarrow 7c - 8ct^2 = t^2 + 3.$$

Collect and factor

$$7c - 3 = (1 + 8c)t^2.$$

Take square roots

$$t^2 = \frac{7c - 3}{1 + 8c} \Rightarrow t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}.$$

Final answer

$$t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}$$

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

Worked solution

January 2020 | Question 16 | 4 marks

Family: Rearrange a subject inside one fraction • Variant: Subject in the denominator

Method / working

1. Square the equation: $y^2 = (x+1)/(x-4)$.
2. Multiply through by $x-4$ to remove the fraction.
3. Expand and collect x terms: $x(y^2-1) = 4y^2+1$.
4. Divide by y^2-1 : $x = (4y^2+1)/(y^2-1)$.

Final answer: $x = (4y^2+1)/(y^2-1)$